



Simon Pelletier

CEP | M.A.Sc.

R&D | Bachelor in Mechanical Engineering

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in SimPelletierPouliot

SimPelletier

Link to Online Resume

About me

I am a mechanical engineering candidate with a specialization in mechanical design. I possess particular expertise in the use of Python and open-source libraries for the design and validation of mechanical structures, which I have applied to my master's project.

My experience includes mechanical design of composite structures, manufacturing processes, tooling and machine element design. Additionally, I have a strong understanding of cable mechanics, which I acquired as part of my master's research.

Skills

Mechanical Design (SolidWorks, Catia, Fusion360)

Programming (Python, Numba, Latex)

Structural Engineering (Ansys APDL, Altair, Fatigue)

Data Analysis (Numpy, Pandas, Sk-learn)

Computer Vision (OpenCV)

Scientific Visualization (VTK, Paraview)

French(Native) English(Professional)
Spanish(Beginner) Solution-Oriented

Experience

Since 2026

AEOLUS Hardware Coordinator

Preformed Line Products (PLP)

- Hardware Development & Validation
- Material & Deployment Coordination
- Technical Documentation & Training

2025 to 2026

AEOLUS Project Manager

Preformed Line Products (PLP)

- Design of New Monitoring Devices
- Deployment Coordination

2022 to 2024

Practical class teacher for MEC528 course

École de Technologie Supérieure (ETS)

- Synthesize concepts related to machine elements and fatigue.
- Carry out classroom and laboratory exercises.

Part-time

2020 to 2022

CEP Project Manager

Centre de développement des composites du Québec (CDCQ)

- Design of tooling and structural composite parts for high-performance and lightweight solutions.

Sum 2019

Mechanical Designer

Rocky Mountain Bicycles

- Design of reliability test benches for electric motor components.
- Design and implementation of an electric motor efficiency bench test.
- Post-processing of test data.
- Development of test procedures.

Internship

Sum-Fal 2017

Mechanical Designer

Centre de développement des composites du Québec (CDCQ)

- Design of tooling and structural composite parts.
- Implementation of a large-capacity FDM 3D printer.
- Design of a nylon in-situ curing machine.

Internship

Education

2022 to 2025

M.A.Sc. candidate in Mechanical Engineering

To École de Technologie Supérieure (ETS)

Comprehensive mechanical modeling of cables under cyclic loading for service life prediction.

- 3D reconstruction of cable internal wire paths from computed tomography data using image processing (OpenCV).
- Macro-scale modeling of internal cable structures using homogenization approaches (Python, APDL).
- Wire-level modeling of a braided structure inside cable outer layers with several million contact points.
- Analysis of contact interaction data using open-source 3D scientific visualization tools (VTK).

Win-Spr 2020

Abroad Studies

At Universitat Politècnica de Catalunya (UPC)

2016 to 2020

B.Sc. in Mechanical Engineering

From École de Technologie Supérieure (ETS)

2013 to 2016

College in Mechanical Engineering

From Cégep Lévis-Lauzon

Projects

Spr 2022

Custom Linear FEA Program of Composite Beams and Plates

École de Technologie Supérieure (ETS-SYS806)

- Orthotropic properties derived from the Classical Laminate Theory.
- Beam element formulation based on Timoshenko beam theory.
- Plate element formulation based on Reissner-Mindlin theory.
- Validated with Ansys® software using the ACP module.

Win-Spr 2020

Design of a Bicycle Cable Disc Brake

Universitat Politècnica de Catalunya (UPC)

2016 to 2018

Participation in the Student Project C-Class Catamaran

École de Technologie Supérieure (ETS)

Spr 2016

Design and Manufacture of a Nylon Rope Machine Cutter

From Cégep Lévis-Lauzon

- Design and manufacture of the prototype.
- Program an Arduino board for system operation and user interface.

Awards

2023

Recipient of an Excellence Scholarship (ETS).

2018 & 2020

Recipient of Two Scholarships from Private Companies to Encourage Career Choice (ETS).

(🔗) - Link Available

References Upon Request